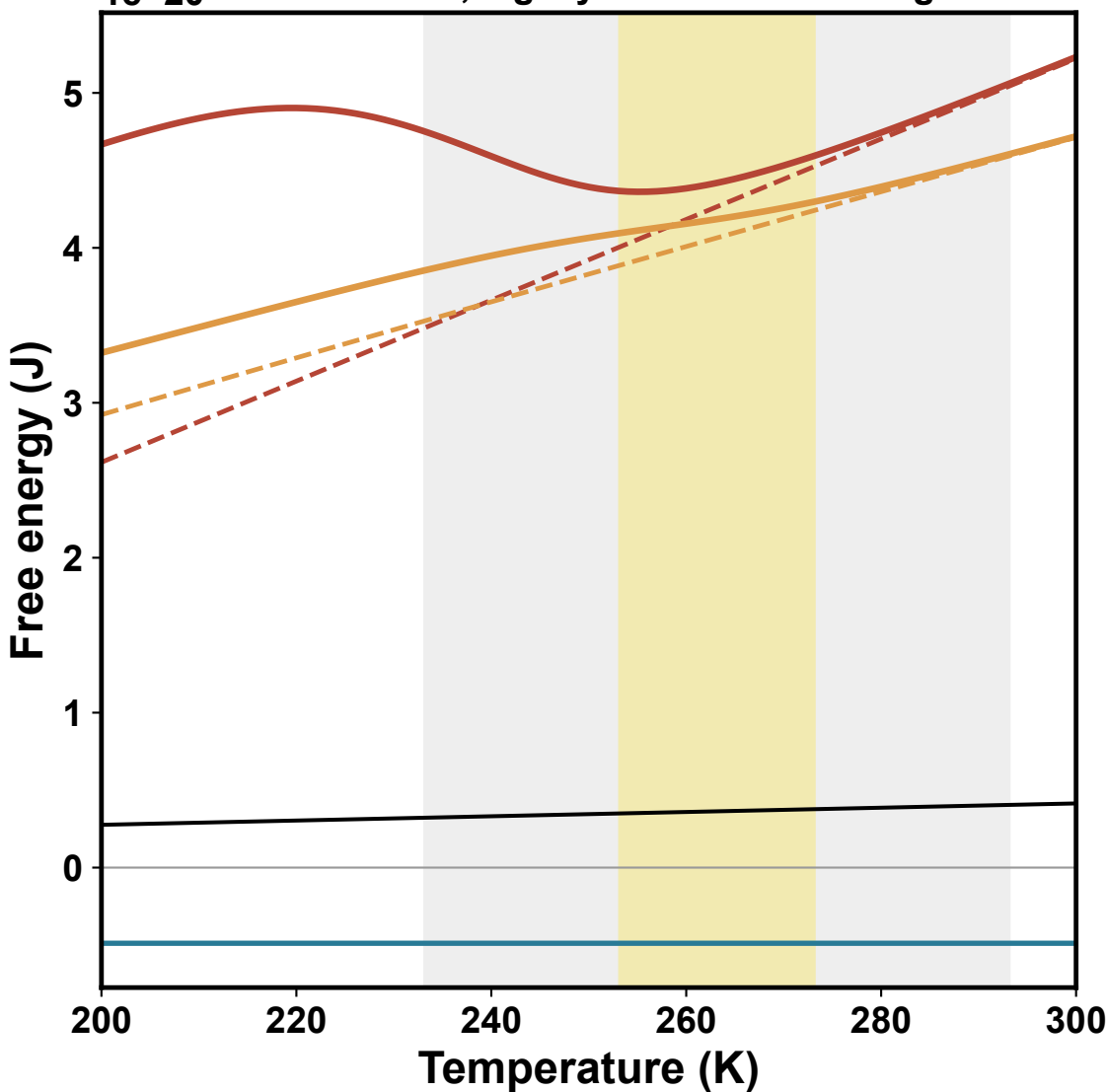


3. Magnetic free energy + rotation entropy

1e-20 20 s window; legacy face twist +/- 5 deg



- face, 20 s
- face, equilibrium
- tip, 20 s
- tip, equilibrium
- face vdW (separate)
- $k_B T$ (reference)
- Experimental range
- Transient aggregation (experiment)